# Chapter # 13
# Climate Change

**Student Learning Outcomes**

**After studying this chapter, students will be able to:**

- [B-12-U-01] Describe how climate change impacts flora and fauna.
- [B-12-U-02] Describe how climate change can impact ocean biology in terms of its temperature and acidity as well as the resulting harmful effects.
- [B-12-U-03] Name species that have gone extinct due to climate change.

Climate change refers to long-term alterations in global or regional climate patterns, primarily driven by natural processes and human activities. While the Earth's climate has naturally fluctuated over geological time, recent decades have seen unprecedented warming, largely due to the increased concentration of greenhouse gases such as carbon dioxide ($CO_2$), methane ($CH_4$), and nitrous oxide ($N_2O$) in the atmosphere. These gases trap heat, creating the "greenhouse effect," which is essential for life but becomes harmful when intensified by excessive emissions.

Understanding climate change builds on concepts from ecology, environmental science, and biology. Organisms adapt to their habitats, but climate change alters temperature, precipitation patterns, and the frequency of extreme events, affecting the availability of resources like water, food, and shelter for both flora and fauna. These changes can lead to shifts in species distribution, altered reproductive cycles, and even extinction.

Oceans, which cover more than 70% of the Earth's surface, are particularly sensitive to climate change. Rising sea temperatures impact marine life by changing habitats, influencing migration patterns, and stressing species that are temperature-sensitive. Ocean acidification, caused by increased $CO_2$ absorption, disrupts calcium carbonate formation in corals and shell-forming organisms, ultimately affecting food webs and human communities dependent on marine resources. Understanding these processes requires recalling prior knowledge of marine biology, cellular processes like respiration and calcification, and the interdependence of species within ecosystems.

Some species have already gone extinct due to climate change, serving as stark reminders of the consequences of environmental disruption. For example, habitat loss due to rising sea levels or melting ice has directly affected small island mammals and polar-adapted species. Recognizing these examples helps highlight the urgency of conservation biology and biodiversity protection.

This content explores the effects of climate change on both terrestrial and marine life, examines how species are impacted, and highlights examples of extinction caused by climate-driven environmental changes. By connecting previous learning with these concepts, students can develop a holistic understanding of climate change and its biological consequences.

**SLO [B-12-U-01]**
**Describe how climate change impacts flora and fauna.**

## IMPACT OF CLIMATE CHANGE ON FLORA AND FAUNA

Climate change profoundly affects both plant (flora) and animal (fauna) life across the globe. Rising temperatures, altered precipitation patterns, and increased frequency of extreme weather events create challenges for organisms that have evolved under specific environmental conditions. These changes can disrupt growth, reproduction, distribution, and survival of species, leading to shifts in ecosystems and biodiversity loss.

studyplusplus.com (study++)

## Impact on Flora (Plants)

### 1. Changes in Distribution

**Figure: Activity to demonstrate the impact of changes in growing seasons.**

Many plant species are highly sensitive to temperature, precipitation, and seasonal patterns. As global temperatures rise, plants adapted to cooler environments are forced to migrate toward higher altitudes or latitudes to survive. For instance, **alpine plants in the European Alps** are moving uphill as lower-altitude areas become too warm. Similarly, **boreal forests in Canada and Russia** are slowly shifting northward. However, mountaintops and northern limits provide finite space, creating "habitat squeeze," which can result in **local extinctions.** Coastal plants, such as **mangroves,** are also affected by rising sea levels, forcing migration inland, which is often limited by human settlements or terrain.

### 2. Altered Growth and Reproduction

Temperature and precipitation directly influence plant growth, flowering, and seed production. Warmer temperatures can accelerate flowering or leafing in some species; for example, **cherry trees in Japan now flower earlier by up to two weeks** compared to 50 years ago. However, if pollinators like bees or butterflies have not yet emerged, seed production may decline. Similarly, **droughts in sub-Saharan Africa** have reduced crop yields of sorghum and millet, while **monsoon flooding in India and Bangladesh** has uprooted rice fields and washed away nutrients, stunting plant growth and threatening food security.

### 3. Phenological Shifts

Phenology refers to the timing of life cycle events. Climate change can cause earlier or delayed leaf emergence, flowering, or seed dispersal. For example, **North American wildflowers are flowering weeks earlier** than historical averages, while some migratory birds, such as the **European pied flycatcher**, arrive too late to feed on emerging insects linked to these plants. This mismatch disrupts food webs and can lead to population declines of both plants and dependent species. Phenological shifts have also been observed in **grapevines in France and Italy**, affecting wine production due to altered ripening periods caused by warmer springs and hotter summers.

### 4. Increased Vulnerability to Pests and Diseases

Higher temperatures, humidity changes, and milder winters create favorable conditions fo r pests, pathogens, and invasive species. For instance, **the mountain pine beetle in North America** has expanded its range due to warmer winters, devastating pine forests. **Coffee plants in Central and South America** are increasingly affected by coffee leaf rust, a fungal disease worsened by rising temperatures and humidity. Invasive plant species, such as **kudzu in the southeastern United States**, also proliferate more easily under warmer, wetter conditions, outcompeting native flora and further destabilizing ecosystems.

## Impact on Fauna (Animals)

### 1. Habitat Loss and Fragmentation

Rising temperatures, melting ice, sea-level rise, desertification, and deforestation all destroy or shrink natural habitats, putting many species at risk. **Polar bears,** for example, rely on sea ice to hunt seals. As ice melts earlier in the season and forms later, they must swim longer distances, expending more energy and reducing hunting success, which threatens survival. **Amphibians,** such as the **golden toad in Costa Rica,** face extinction when their freshwater wetlands dry up due to higher temperatures and altered rainfall patterns. Coastal habitats like **mangroves** and **coral reefs** are also lost to sea-level rise and ocean warming, which removes shelter and breeding grounds for countless fish, crustaceans, and birds. Fragmented forests in the Amazon and Southeast Asia have confined species like **orangutans and jaguars** to isolated patches, limiting their ability to find food and mates.


**Figure: Impact of climate change on Polar Bear' habitat.**

2. **Altered Distribution and Migration**

Animals often respond to changing climates by moving toward cooler regions or deeper waters. For example, **monarch butterflies** have shifted their migratory patterns in North America due to warmer temperatures and changing precipitation. **Atlantic cod and other fish species** are moving to deeper, colder waters in response to ocean warming. Birds like the **barn swallow in Europe** now arrive earlier in spring or shift their breeding grounds northward. However, not all species can migrate. Limited dispersal abilities, geographical barriers such as mountains or urban areas, and habitat fragmentation prevent many species from reaching suitable environments, putting them at higher risk of population decline.

3. **Physiological Stress**

Temperature increases can exceed the tolerance limits of many animals, leading to heat stress, reduced reproduction, or death. **Coral reefs** are highly sensitive to even small increases in water temperature; prolonged warming causes coral bleaching, where corals expel the symbiotic algae that provide nutrients, often resulting in mass mortality. **Amphibians** like the **fire salamander** suffer dehydration and reduced reproductive success under higher temperatures. Similarly, **polar and high-altitude mammals** such as snow leopards experience stress when warming reduces their prey availability and snow-covered habitat.

4. **Disruption of Food Webs**

Climate change affects the timing of food availability and the interactions between predators and prey. For instance, if flowering plants bloom earlier due to warmer springs, **pollinators such as bees and butterflies** may emerge too late, reducing their survival and reproduction. Fish species, such as **Atlantic salmon**, face challenges if their prey move to cooler waters earlier than usual. Predators like **arctic foxes** and **lynx** may experience food shortages if small mammals like lemmings and hares decline due to snow cover loss or habitat shifts. Even large mammals like **African elephants** can be indirectly affected when prolonged drought reduces fruiting trees, forcing longer migrations for food.

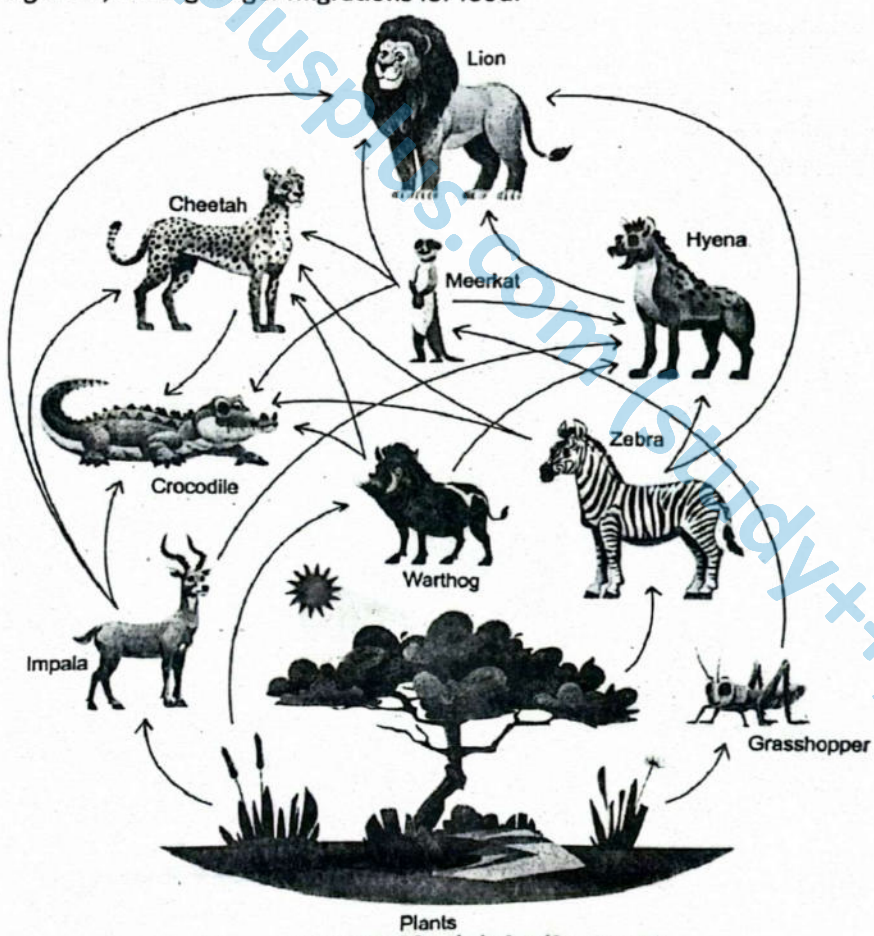


Figure: Network of food chains (food web)

studyplusplus.com (study++)

## Ecosystem-Level Effects

### 1. Biodiversity Loss

Climate change threatens biodiversity by creating conditions that many species cannot tolerate or adapt to quickly enough. Species unable to migrate or adjust to rising temperatures, altered rainfall, or extreme events face local or global extinction. For example, the **Bramble Cay melomys**, a small rodent from a low-lying island in the Great Barrier Reef, became the first mammal confirmed extinct due to rising sea levels and habitat inundation. Similarly, **coral species on the Great Barrier Reef** have suffered mass die-offs from repeated bleaching events caused by elevated sea temperatures. Loss of biodiversity reduces ecosystem resilience, meaning ecosystems are less able to recover from disturbances such as storms, fires, or disease outbreaks.

### 2. Invasive Species Expansion

Warmer temperatures and altered precipitation patterns often favor opportunistic invasive species, which thrive under new conditions and outcompete native species. For instance, **kudzu** in the southeastern United States spreads rapidly with warmer winters, choking native plants. In marine systems, **lionfish** have expanded in the Caribbean and Atlantic waters due to rising sea temperatures, preying on native fish and altering reef community structures. Invasive pests, such as the **pine beetle in North American forests**, reproduce more effectively during milder winters, devastating native pine species and transforming forest ecosystems. These invasions reduce native biodiversity and destabilize ecosystem function.

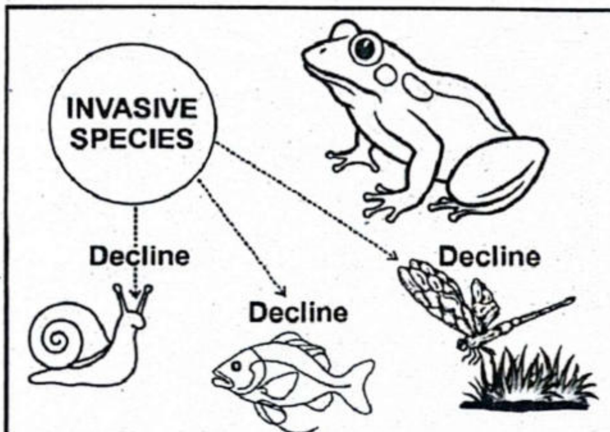

**Figure: Effect of invasive species on other species.**

### 3. Altered Ecosystem Services

Climate-induced changes in plant and animal populations directly affect the ecosystem services humans rely on. **Pollination services** decline when bees and other pollinators emerge at mismatched times relative to flowering plants, reducing crop yields. **Seed dispersal and forest regeneration** are affected when frugivorous birds or mammals decline or shift ranges. **Carbon sequestration** decreases when large forests are degraded or lost, such as the Amazon rainforest experiencing dieback during prolonged droughts. Soil stabilization is compromised when vegetation is lost to desertification or extreme storms, increasing erosion and flooding risks. These disruptions not only threaten biodiversity but also have tangible consequences for agriculture, fisheries, water quality, and climate regulation.

**SLO [B-12-U-02]**
**Describe how climate change can impact ocean biology in terms of its temperature and acidity as well as the resulting harmful effects.**

## CLIMATE CHANGE IMPACTS ON OCEAN ECOSYSTEM

Climate change significantly alters oceanic systems, affecting both their physical properties and biological dynamics. Oceans play a critical role in regulating Earth's climate by absorbing heat and carbon dioxide. However, rising greenhouse gas emissions are pushing ocean ecosystems beyond their natural capacity to maintain equilibrium.

The primary impacts of climate change on marine environments include:

- Rising ocean temperatures
- Ocean acidification (decline in pH)
- Alterations in marine biodiversity, species distribution, and ecosystem functioning

studyplusplus.com (study++)

## Climate Change impacts Ocean Temperature

Climate change impacts ocean temperatures through several human and natural factors:

### Global Warming and the Greenhouse Gasses

Ocean temperatures are increasing as a direct result of global warming, which is driven by the accumulation of greenhouse gases such as carbon dioxide ($CO_2$), methane ($CH_4$), and nitrous oxide ($N_2O$) in the atmosphere. The warming of ocean waters has both direct and indirect effects on marine ecosystems and global climate systems.

### Greenhouse Gas Accumulation

Oceans absorb more than 90% of the excess heat trapped by greenhouse gases. This massive heat uptake moderates surface temperatures on land but leads to thermal stress for marine organisms and disrupts ocean circulation patterns.

**SCIENCE TITBITS**

Warmer water expands, a process known as thermal expansion, which contributes to rising sea levels. This process occurs as water molecules move further apart when they are heated, leading to the expansion of the ocean's volume.

**How can you prove thermal expansion in the laboratory?**

### Thermal Expansion

As water is heated, its molecules move further apart, causing it to expand. This process—known as **thermal expansion**—contributes significantly to sea level rise, even in the absence of melting ice caps.

**Thermal expansion** can be demonstrated in the laboratory using the following setup:

- Fill a glass flask with colored water.
- Insert a thin glass tube into the flask and seal the top to prevent evaporation.
- Mark the initial water level in the tube.
- Gently heat the flask using warm water or a lamp.
- Observe the rise in water level in the tube as the water warms and expands.

### High heat capacity of ocean

Water has an exceptionally high specific heat capacity, allowing the oceans to absorb and store vast quantities of thermal energy with only slight changes in temperature. While this property buffers the Earth's climate, it results in the long-term accumulation of heat in the oceans, affecting marine life.

### Large Global Surface Area

Covering more than 70% of the planet's surface, oceans receive the majority of solar radiation and heat exchange with the atmosphere. This makes them a key component of Earth's heat balance but also highly vulnerable to prolonged warming.

### Ocean Circulation and Heat Redistribution

Major currents such as the **Gulf Stream** and **Atlantic Meridional Overturning Circulation (AMOC)** help distribute heat around the globe. Climate change is weakening these currents due to alterations in salinity and temperature, disrupting the normal distribution of ocean heat and affecting regional climates.

**El Niño and La Niña Events:**

These periodic climatic events in the Pacific Ocean affect global weather and ocean temperatures.

- **El Niño** brings unusually warm surface waters to the eastern Pacific, elevating ocean temperatures and influencing storm patterns.
- **La Niña** causes cooler-than-normal waters but can set the stage for more intense warming events in subsequent years.

**Melting of Sea Ice**

Rising ocean temperatures accelerate the melting of sea ice in polar regions. Sea ice acts as a reflective *barrier (high albedo)* that bounces solar radiation back into space. When it melts, darker ocean water absorbs more heat, reinforcing the warming trend—a feedback mechanism that intensifies climate change.

studyplusplus.com (study++)

32

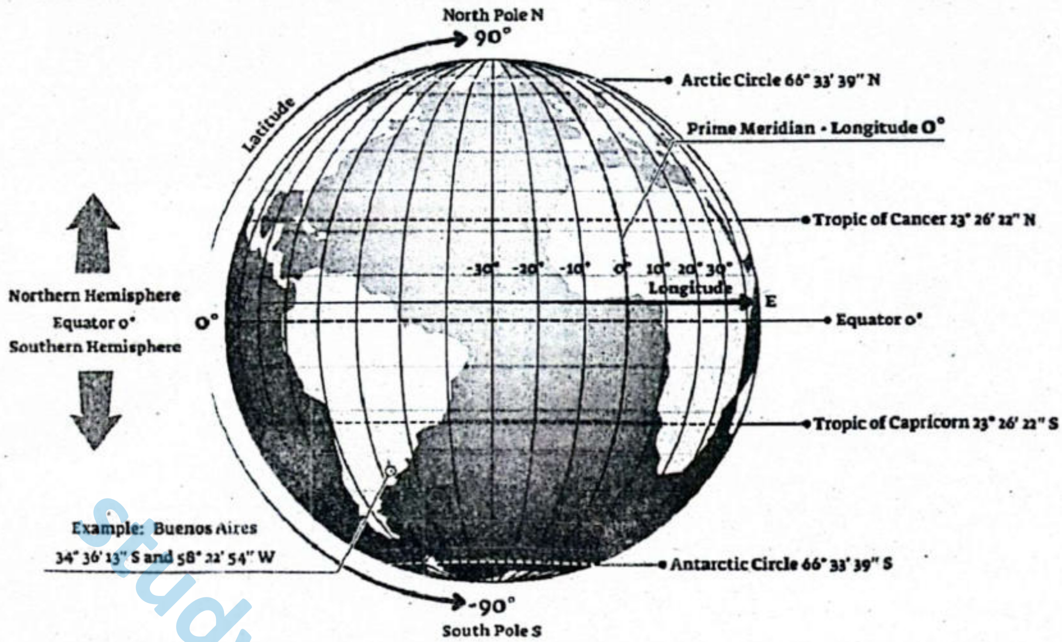


**Figure:** Geographical distribution of earth. The numbers like 23°26′22″N, represent the geographic coordinates that specify the exact latitude. For example: 23°26′22″N is the precise position of the Tropic of Cancer, which is one of the major circles of latitude on Earth.

**23°:** This represents 23 degrees of latitude north of the equator.

**26':** The single apostrophe (') represents 26 minutes of latitude. There are 60 minutes in one degree.

**22":** The double apostrophe (") represents 22 seconds of latitude. There are 60 seconds in one minute.

**N:** Indicates that this latitude is in the Northern Hemisphere.

**Meaning:** The Tropic of Cancer is located at 23 degrees, 26 minutes, and 22 seconds north of the equator.

## Climate Change impacts Ocean Acidification (Low pH)

Ocean acidification refers to the reduction in seawater pH caused by the uptake of atmospheric $CO_2$. This process has serious implications for marine organisms, especially those that form calcium carbonate shells or skeletons.

In climate science, a *positive feedback loop* is a process that **amplifies** or **reinforces** the original change. In this case:

- Melting ice reduces Earth's albedo (reflectivity), so more solar energy is absorbed by the ocean.
- Warmer oceans store **less carbon dioxide**, releasing more $CO_2$ into the atmosphere.
- Higher atmospheric $CO_2$ levels trap more heat, which leads to further ocean warming and more ice melt.

Each step in this cycle **intensifies** the next, making the initial warming worse—hence, it's a **positive feedback**, even though its consequences are detrimental.

**CRITICAL THINKING**

Climate change creates feedback loops that amplify ocean warming. For example, as ice melts and more heat is absorbed by the ocean, the ocean's ability to store carbon dioxide diminishes. This, in turn, increases atmospheric $CO_2$ levels, trapping more heat and further warming both the atmosphere and the oceans.

Is this feedback positive or negative?

**Committed warming** refers to the continued warming of Earth even if greenhouse gas emissions stop today. This happens because:

- **Oceans have high heat capacity** and have already absorbed most of the excess heat, which is slowly released over time.
- **Carbon dioxide stays in the atmosphere** for centuries, continuing to trap heat.
- **The climate system responds slowly**, so warming continues as it adjusts to past emissions.

studyplusplus.com (study++)

Thus, due to heat already stored in the oceans and persistent $CO_2$, the planet will keep warming for decades to centuries.

- **Ocean as a Carbon Sink**

The ocean absorbs roughly 30% of human-generated $CO_2$. This capacity buffers atmospheric $CO_2$ levels but alters the chemical balance of seawater.

> **CRITICAL THINKING**
>
> Even if greenhouse gas emissions were to stop today, the heat already absorbed by the oceans would continue to warm the planet for centuries. This is known as committed warming.
> Why does this happen?

- **$CO_2$ Dissolving in Ocean Water**

When $CO_2$ dissolves in seawater, it undergoes a chemical reaction to form carbonic acid ($H_2CO_3$):

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

Carbonic acid is unstable and dissociates into bicarbonate ions ($HCO_3^-$) and hydrogen ions ($H^+$):

$$H_2CO_3 \longrightarrow HCO_3^- + H^+$$

The increase in hydrogen ions ($H^+$) is what causes the pH of the ocean to decrease, making the ocean more acidic. Ocean acidification refers to this long-term lowering of pH levels due to the absorption of $CO_2$. Ocean acidification and climate change are interlinked. As climate change causes ocean temperatures to rise, it reduces the ability of the ocean to absorb $CO_2$. Warmer water holds less gas, meaning that in the future, more $CO_2$ will remain in the atmosphere, accelerating global warming. As the ocean becomes more acidic, it affects the delicate balance of ocean chemistry, which has serious consequences for marine life and ecosystems.

- **Regional Variability**

Polar oceans are particularly susceptible to acidification because cold water absorbs $CO_2$ more readily. This threatens organisms adapted to these environments, such as **cold-water corals**, which are essential for biodiversity in high-latitude marine ecosystems.

## Harmful Effects of High Ocean Temperature and Ocean Acidification on Marine Biology

High ocean temperatures and ocean acidification, both consequences of climate change, have profound and harmful effects on marine biology. These environmental changes threaten the survival of numerous marine species and ecosystems, disrupting food webs, biodiversity, and the services oceans provide to humanity.

- **Harmful Effects of High Ocean Temperature on Marine Biology**

High Ocean Temperature affects marine life in several ways:

**Coral bleaching:** Coral reefs, among the most diverse marine ecosystems, rely on symbiotic algae (zooxanthellae) for energy. Elevated water temperatures cause corals to expel these algae, leading to **bleaching**—a whitening of corals that signals extreme stress. Prolonged bleaching can result in coral death, disrupting the entire reef ecosystem.

**Ecosystem Disruption**: Temperature-sensitive species, including plankton and certain fish, may migrate toward cooler waters. Such shifts can destabilize marine food webs and reduce local fishery yields, particularly in tropical and subtropical regions.

**Reduced Reproductive Success**: Many marine species are ectothermic (cold-blooded), so their metabolism and reproductive cycles are highly dependent on temperature. Warmer waters can lead to:

- Altered spawning seasons
- Smaller body size at maturity
- Lower reproductive output

**Marine Heatwaves**: Short-term yet intense rises in sea temperature, known as marine heatwaves, are becoming more frequent. These events cause mass mortality in fish, mollusks, corals, and marine mammals, with lasting effects on ecosystem structure and services.

**Polar Ice Melt and Habitat Loss**: Species such as polar bears, walruses, and seals depend on sea ice for hunting, resting, and breeding. The rapid loss of sea ice limits access to critical resources, contributing to declining populations and increased stress on Arctic and Antarctic ecosystems.

***Ocean Deoxygenation:*** *Warmer* oceans retain less dissolved oxygen. This leads to the formation *of* **hypoxic zones** (*low* oxygen) and **anoxic zones** (no oxygen), where marine life cannot survive. Deoxygenation reduces biodiversity, alters predator-prey relationships, and negatively impacts global fisheries.

studyplusplus.com (study++)

Thus, due to heat already stored in the oceans and persistent $CO_2$, the planet will keep warming for decades to centuries.

- **Ocean as a Carbon Sink**

The ocean absorbs roughly 30% of human-generated $CO_2$. This capacity buffers atmospheric $CO_2$ levels but alters the chemical balance of seawater.

> **CRITICAL THINKING**
>
> Even if greenhouse gas emissions were to stop today, the heat already absorbed by the oceans would continue to warm the planet for centuries. This is known as committed warming.
> Why does this happen?

- **$CO_2$ Dissolving in Ocean Water**

When $CO_2$ dissolves in seawater, it undergoes a chemical reaction to form carbonic acid ($H_2CO_3$):

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

Carbonic acid is unstable and dissociates into bicarbonate ions ($HCO_3^-$) and hydrogen ions ($H^+$):

$$H_2CO_3 \longrightarrow HCO_3^- + H^+$$

The increase in hydrogen ions ($H^+$) is what causes the pH of the ocean to decrease, making the ocean more acidic. Ocean acidification refers to this long-term lowering of pH levels due to the absorption of $CO_2$. Ocean acidification and climate change are interlinked. As climate change causes ocean temperatures to rise, it reduces the ability of the ocean to absorb $CO_2$. Warmer water holds less gas, meaning that in the future, more $CO_2$ will remain in the atmosphere, accelerating global warming. As the ocean becomes more acidic, it affects the delicate balance of ocean chemistry, which has serious consequences for marine life and ecosystems.

- **Regional Variability**

Polar oceans are particularly susceptible to acidification because cold water absorbs $CO_2$ more readily. This threatens organisms adapted to these environments, such as **cold-water corals**, which are essential for biodiversity in high-latitude marine ecosystems.

## Harmful Effects of High Ocean Temperature and Ocean Acidification on Marine Biology

High ocean temperatures and ocean acidification, both consequences of climate change, have profound and harmful effects on marine biology. These environmental changes threaten the survival of numerous marine species and ecosystems, disrupting food webs, biodiversity, and the services oceans provide to humanity.

- **Harmful Effects of High Ocean Temperature on Marine Biology**

High Ocean Temperature affects marine life in several ways:

**Coral bleaching:** Coral reefs, among the most diverse marine ecosystems, rely on symbiotic algae (zooxanthellae) for energy. Elevated water temperatures cause corals to expel these algae, leading to **bleaching**—a whitening of corals that signals extreme stress. Prolonged bleaching can result in coral death, disrupting the entire reef ecosystem.

**Ecosystem Disruption**: Temperature-sensitive species, including plankton and certain fish, may migrate toward cooler waters. Such shifts can destabilize marine food webs and reduce local fishery yields, particularly in tropical and subtropical regions.

**Reduced Reproductive Success**: Many marine species are ectothermic (cold-blooded), so their metabolism and reproductive cycles are highly dependent on temperature. Warmer waters can lead to:

- Altered spawning seasons
- Smaller body size at maturity
- Lower reproductive output

**Marine Heatwaves**: Short-term yet intense rises in sea temperature, known as marine heatwaves, are becoming more frequent. These events cause mass mortality in fish, mollusks, corals, and marine mammals, with lasting effects on ecosystem structure and services.

**Polar Ice Melt and Habitat Loss**: Species such as polar bears, walruses, and seals depend on sea ice for hunting, resting, and breeding. The rapid loss of sea ice limits access to critical resources, contributing to declining populations and increased stress on Arctic and Antarctic ecosystems.

***Ocean Deoxygenation:*** Warmer oceans retain less dissolved oxygen. This leads to the formation of **hypoxic zones** (low oxygen) and **anoxic zones** (no oxygen), where marine life cannot survive. Deoxygenation reduces biodiversity, alters predator-prey relationships, and negatively impacts global fisheries.

DO YOU KNOW?

The Intergovernmental Panel on Climate Change (IPCC) is an international body established in 1988 by the United Nations Environment Programme (UNEP) and the World Meteorological Organization (WMO). Its primary role is to assess scientific research related to climate change, including its causes, impacts, and potential strategies for adaptation and mitigation. The IPCC serves as a bridge between the scientific community and policymakers, ensuring that climate actions are based on the best available evidence.

**SLO [B-12-U-03]**
**Name species that have gone extinct due to climate change.**

## SPECIES EXTINCTION DUE TO CLIMATE CHANGE

Climate change is a significant driver of species extinction, as it rapidly alters the environmental conditions necessary for the survival of various organisms. Shifts in global temperature, altered precipitation patterns, sea level rise, and the increased frequency and intensity of extreme weather events can exceed a species' physiological tolerance and capacity to adapt. When habitat loss, resource scarcity, and ecological imbalances occur faster than evolutionary or behavioral adjustments, species may face extinction.

### Documented Species Extinctions Linked to Climate Change

Several species have already been driven to extinction, in whole or in part, due to climate change. These documented cases highlight the ecological consequences of global warming on vulnerable flora and fauna.

- **Bramble Cay Melomys (*Melomys rubicola*)**

  This small rodent, once endemic to Bramble Cay—a low-lying island in the Torres Strait, northeast of Australia—was officially declared extinct in 2016. It is recognized as the first recorded mammal extinction directly attributed to anthropogenic climate change. Rising sea levels and repeated storm surges inundated the island, destroying the vegetation the Melomys relied upon for food and shelter.

- **Golden Toad (Incilius periglenes)**

  The golden toad, native to the Monteverde Cloud Forest in Costa Rica, was last seen in 1989 and is now presumed extinct. Climate change-induced alterations in the region's humidity and temperature patterns are believed to have significantly reduced breeding habitat suitability. These changes also created favorable conditions for the spread of chytridiomycosis, a fungal disease that further accelerated the population decline.

- **Pinta Island Tortoise (*Chelonoidis abingdonii*)**

  Native to Ecuador's Galápagos Islands, the Pinta Island tortoise population was already dwindling due to overexploitation and habitat degradation. The death of the last known individual, *Lonesome George*, in 2012 marked the species' extinction. While historical human activity was the primary cause, climate change contributed by altering the island's vegetation and water availability, compounding existing environmental stresses.

- **Atelopus Frogs (Harlequin Frogs, *Atelopus* spp.)**

  Numerous species within the genus *Atelopus*, distributed across Central and South America, have suffered dramatic population declines and extinctions in recent decades. A major factor in their decline is the chytrid fungus (*Batrachochytrium dendrobatidis*), whose emergence and spread have been strongly linked to changing climate patterns—especially rising temperatures and fluctuating humidity levels in mountainous cloud forests. These amphibians are highly sensitive to microclimatic changes, making them especially vulnerable.

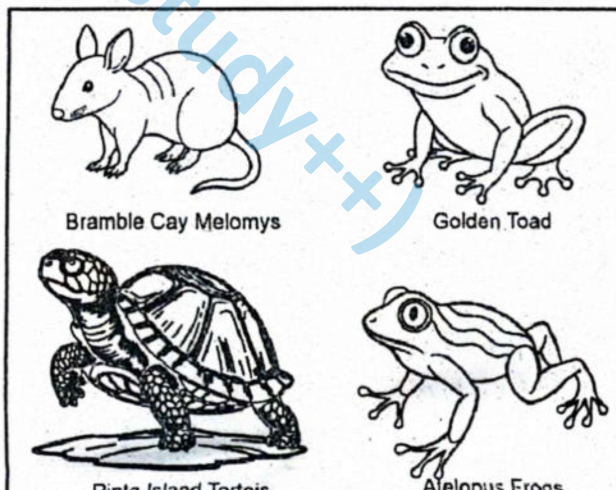

**Figure: Species that have gone extinct due to climate change**

studyplusplus.com (study++)

## Species that have gone endangered due to climate change (Extra Reading Material)

While several species have gone extinct, many species have become endangered due to the impacts of climate change. Here are some notable examples:

- **Polar Bears (Ursus maritimus):** Polar bears rely on sea ice to hunt for seals. As Arctic ice melts due to rising temperatures, their hunting grounds shrink, threatening their survival.
- **Adélie Penguins (Pygoscelis adeliae):** These penguins depend on Antarctic Sea ice for breeding and feeding. Warming temperatures and changing ice patterns are reducing their habitat.
- **Koalas (Phascolarctos cinereus):** Increased heatwaves and bushfires, exacerbated by climate change, along with habitat loss, have pushed koalas closer to endangerment.
- **Snow** Leopards **(Panthera uncia):** Rising temperatures are affecting the mountain ecosystems they inhabit, shrinking their prey base and increasing human-wildlife conflict.
- **Coral** Species **(e.g., Staghorn Coral):** Coral reefs are being devastated by ocean warming and acidification, leading to large-scale coral bleaching events that threaten the survival of many coral species.
- Monarch **Butterflies (Danaus plexippus):** Climate change is altering their migration patterns and reducing the availability of milkweed, their primary food source, pushing them toward endangerment.
- **Atlantic Cod (Gadus morhua):** Rising ocean temperatures are disrupting cod breeding grounds and food availability, causing declines in their population.

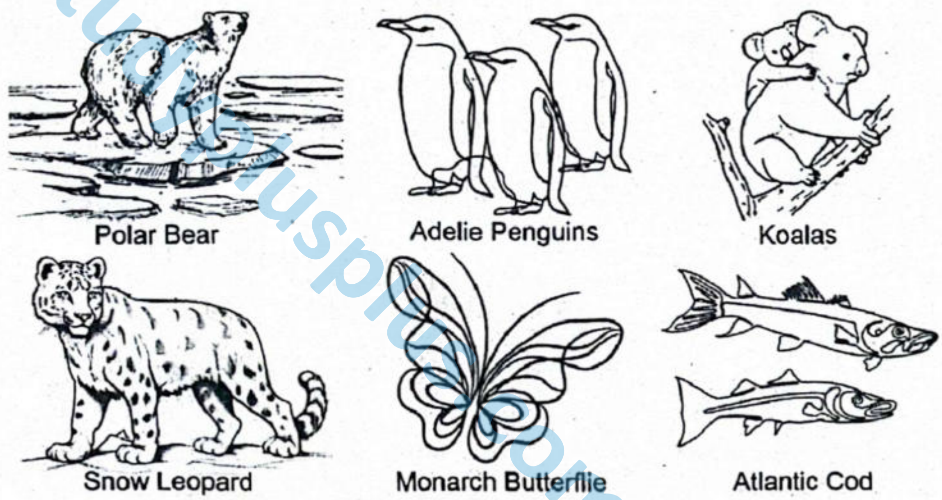


**Figure: Species that have gone endangered due to climate change**

These species are just a few of the many that are struggling to adapt to the rapidly changing climate.

### SLOs Based (MCQs)

**1. Which gas is primarily responsible for ocean acidification?**
A. Oxygen B. Methane
C. Carbon dioxide D. Nitrogen

**2. What percentage of excess heat from greenhouse gases is absorbed by oceans?**
A. 50% B. 70%
C. 90% D. 30%

**3. Which process contributes to sea level rise due to warming oceans?**
A. Evaporation B. Condensation
C. Thermal expansion D. Ice sublimation

**4. Oceans absorb large amounts of heat due to their:**
A. Low salinity B. High density
C. High heat capacity D. Low surface tension

**5. What is the effect of rising sea temperatures on marine circulation?**
A. Increases current speed
B. Strengthens ocean gyres
C. Disrupts circulation patterns
D. Improves nutrient cycling

**6. Which current is most associated with climate regulation in the Atlantic Ocean?**
A. Humboldt Current B. Gulf Stream
C. Canary Current
D. Indian Monsoon Current

**7. The El Niño phenomenon results in:**
A. Global cooling
B. Reduced rainfall
C. Ocean warming in the Pacific
D. Stronger polar vortex

studyplusplus.com (study++)

**8. La Niña is characterized by:**
A. Intense ocean acidification
B. Ocean cooling in the Pacific
C. Weakening of the ozone layer
D. Increase in volcanic activity

**9. Melting sea ice reduces Earth's albedo by:**
A. Reflecting more sunlight
B. Absorbing more heat
C. Increasing cloud formation
D. Releasing more ozone

**10. Which condition arises from low oxygen in ocean water?**
A. Hyperoxia B. Eutrophication
C. Deoxygenation D. Thermocline shift

**11. Which marine organisms are directly threatened by lower pH in oceans?**
A. Jellyfish B. Sharks
C. Calcifying organisms D. Cephalopods

**12. Which equation correctly represents carbonic acid formation in seawater?**
A. $CO_2 + H_2O \rightarrow H_2CO_3$ B. $CO_2 + O_2 \rightarrow CO_4$
C. $CO_2 \rightarrow CO + O_2$ D. $H_2O \rightarrow OH^- + H^+$

**13. Ocean acidification results in an increase in:**
A. Hydroxide ions B. Carbon monoxide
C. Hydrogen ions D. Ammonia

**14. A consequence of coral bleaching is:**
A. Increase in chlorophyll
B. Loss of symbiotic algae
C. Hardening of coral
D. Increase in carbon storage

**15. Marine heatwaves primarily result in:**
A. Enhanced fish growth
B. Mass mortality of marine organisms
C. Acid rain formation
D. Ocean desalination

**16. Oceans act as carbon sinks by absorbing approximately:**
A. 10% of $CO_2$
B. 30% of anthropogenic $CO_2$
C. 60% of $CO_2$ emissions
D. 5% of global carbon

**17. Why are polar regions more vulnerable to ocean acidification?**
A. Higher temperatures B. More carbonates
C. Colder waters absorb more $CO_2$
D. Less salt in water

**18. Which statement best defines committed warming?**
A. Ongoing warming caused by future emissions
B. Warming from volcanic eruptions
C. Future warming from existing heat in oceans
D. Cooling due to ice melt

**19. Which laboratory setup demonstrates thermal expansion of water?**
A. Heat and stir a metal rod
B. Freeze water in a beaker
C. Heat colored water in a sealed tube
D. Use a vacuum chamber

**20. Deoxygenated zones in oceans are known as:**
A. Photosynthetic layers B. Thermoclines
C. Anoxic zones D. Hadal zones

**21. Which mammal was the first confirmed extinction due to climate change?**
A. Pinta Island Tortoise B. Golden Toad
C. Bramble Cay Melomys D. Atelopus zeteki

**22. What led to the extinction of Bramble Cay Melomys?**
A. Wildfire B. Disease
C. Sea level rise and storm surges
D. Overhunting

**23. Which extinct species lived in Costa Rica's cloud forests?**
A. Bramble Cay Melomys B. Golden Toad
C. Arctic Fox D. Atelopus varius

**24. What climate factor contributed to the golden toad's extinction?**
A. Freezing conditions B. Prolonged drought
C. Ocean acidification D. Saltwater intrusion

**25. Lonesome George was the last of which species?**
A. Komodo Dragon B. Bramble Cay Melomys
C. Pinta Island Tortoise D. Harlequin Frog

**26. What role did climate change play in the extinction of Lonesome George's species?**
A. Spread of invasive fungi
B. Algal bloom toxicity
C. Disruption of Galápagos ecosystem
D. Ocean salinity loss

**27. Atelopus frogs are mainly affected by:**
A. Water pollution B. Noise pollution
C. Chytrid fungus spread by climate change
D. Genetic disorders

**28. Which environmental condition favors the chytrid fungus?**
A. High salinity B. Stable temperatures
C. Warm and moist conditions
D. Freezing climates

studyplusplus.com (study++)

29. **What common trait do many climate-threatened amphibians share?**
A. Scaly skin
B. Dependence on constant humidity
C. Long lifespan D. Large size

30. **Climate change increases extinction risk by:**
A. Making habitats more diverse
B. Slowing down migration
C. Creating rapid, unadaptable changes
D. Improving food quality

31. **What is true about polar habitat loss?**
A. Melting ice threatens species like walruses and seals
B. Melting ice increases salinity
C. More oxygen dissolves in warm water
D. Low pH increases shell strength

32. **What is true about polar habitat loss?**
A. Arctic habitats are expanding
B. Melting ice threatens species like walruses and seals
C. Low pH increases shell strength
D. Golden Toad thrived in dry habitats

33. **What is true about ocean feedback loops?**
A. Melting ice increases salinity
B. Golden Toad thrived in dry habitats
C. $CO_2$ increases salinity
D. Melting ice reduces albedo, increasing ocean warming

34. **What is true about committed warming?**
A. $CO_2$ increases salinity
B. It refers to ongoing warming due to stored ocean heat
C. Oceans reflect 90% of solar radiation
D. More oxygen dissolves in warm water

35. **What is true about thermal expansion?**
A. Melting ice increases salinity
B. It contributes to sea level rise as water warms
C. Low pH increases shell strength
D. Golden Toad thrived in dry habitats

## Answer Key

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. C | 2. C | 3. C | 4. C | 5. C | 6. B | 7. C | 8. B | 9. B | 10. C | 11. C | 12. A |
| 13. C | 14. B | 15. B | 16. B | 17. C | 18. C | 19. C | 20. C | 21. C | 22. C | 23. B | 24. B |
| 25. C. | 26. C | 27. C. | 28. C | 29. B | 30. C. | 31. A | 32. B | 33. D | 34. B | 35. B | |

## Short Questions

**Q.1 How does climate change contribute to species extinction?**

**Ans.** Climate change contributes to species extinction by altering temperature, rainfall patterns, and the frequency of extreme weather events, which disrupt the habitats that species depend on. Rapid environmental changes can exceed the ability of species to adapt, causing declines in population size and reproductive success. Changes in temperature and precipitation can shift vegetation, water availability, and prey-predator dynamics, further stressing species. Extreme events such as floods, droughts, and storms can destroy critical habitats. As a result, species may face reduced food and shelter availability, increased competition, and heightened vulnerability to disease, ultimately leading to population decline or extinction if adaptive or migratory responses are insufficient.

**MARKING SCHEME (3 Marks)**
- Definition – 1 mark, effects – 2 marks

**Q.2 Describe how the Bramble Cay Melomys became extinct.**

**Ans.** The Bramble Cay Melomys was a small rodent native to Bramble Cay, a tiny island near the northern tip of Australia. Rising sea levels caused by climate change, along with frequent storm surges, led to flooding of its habitat. This destroyed the vegetation and shelter it relied on for food and protection. The limited size of the island meant the species could not migrate or find alternative habitats. These combined pressures ultimately caused the population to collapse. In 2016, the Bramble Cay Melomys was declared extinct, making it the first mammal species to be driven to extinction primarily due to the impacts of climate change, highlighting the vulnerability of island species to environmental shifts.

**MARKING SCHEME (3 Marks)**
- Habitat destruction – 1 mark, effects of flooding – 1 mark, extinction outcome – 1 mark

***Q.3 What were the causes behind the extinction of the Golden Toad?***

**MARKING SCHEME (3 Marks)**
- Climate change effects – 1.5 marks, disease and habitat degradation – 1.5 marks

**Ans.** The Golden Toad, once found in the cloud forests of Costa Rica, disappeared in the late 1980s. Climate change altered the temperature and moisture levels of its high-elevation habitat, creating unsuitable

studyplusplus.com

conditions for breeding and survival. These environmental changes also made the species more susceptible to diseases, particularly chytridiomycosis, caused by a pathogenic fungus. Habitat degradation from altered rainfall and cloud cover reduced available resources. The combination of climate-induced stress and disease outbreaks led to a rapid decline in the Golden Toad population, ultimately resulting in extinction. The case of the Golden Toad demonstrates how climate change can interact with biological factors, such as disease susceptibility, to drive species loss.

**Q.4 Explain the role of climate change in the extinction of the Pinta Island Tortoise.**

**Ans.** The Pinta Island Tortoise, native to the Galápagos Islands, faced multiple threats including habitat destruction and human exploitation. Climate change added additional pressures by disturbing the delicate Galápagos ecosystem, altering rainfall and vegetation patterns. These changes reduced the availability of food and water sources, making survival increasingly difficult for the remaining individuals. The combination of climate-driven habitat stress and other anthropogenic pressures contributed to the decline of the species, culminating in the death of its last known individual, Lonesome George, in 2012. This example highlights how climate change can exacerbate existing threats and accelerate the extinction of species already vulnerable due to limited range and human activities.

> **MARKING SCHEME (3 Marks)**
> - Climate change effect – 1.5 marks, contribution to extinction – 1.5 marks

**Q.5 How does climate change help the spread of chytrid fungus in amphibians?**

**Ans.** Climate change facilitates the spread of chytrid fungus by altering global temperatures and humidity patterns, creating conditions favorable for the pathogen to thrive in new environments. Previously isolated highland habitats become suitable for fungal growth, exposing amphibian populations that have no natural resistance. This has caused massive declines and local extinctions, particularly among Atelopus frog species in Central and South America. The fungus attacks the skin of amphibians, disrupting respiration and water balance, often leading to death. By shifting environmental conditions, climate change indirectly increases the prevalence and impact of chytrid fungus, making it a major driver of amphibian biodiversity loss worldwide.

> **MARKING SCHEME (3 Marks)**
> - Climate change effect on chytrid fungus – 2 marks, effects of spread – 1 mark

**Q.6 What is a climate feedback loop involving ocean warming?**

**Ans.** A climate feedback loop involving ocean warming occurs when rising ocean temperatures amplify global warming in a self-reinforcing cycle. As the oceans warm, polar ice melts, reducing Earth's albedo, which means less sunlight is reflected and more is absorbed by the oceans. Warmer water also holds less carbon dioxide, releasing more $CO_2$ into the atmosphere, which further enhances greenhouse effects. This positive feedback loop accelerates global temperature rise, affecting weather patterns, sea levels, and ecosystems. By continuously reinforcing itself, the loop illustrates how ocean warming can intensify climate change beyond the initial causes, making mitigation increasingly challenging without reducing greenhouse gas emissions.

> **MARKING SCHEME (3 Marks)**
> - Effects of ocean warming – 1 mark, feedback mechanism – 2 marks

**Q.7 Define committed warming.**

**Ans.** Committed warming is the unavoidable future increase in global temperatures caused by greenhouse gases that have already been emitted. Even if all emissions were to stop immediately, the heat stored in the oceans would gradually be released over decades, continuing to warm the planet. This delayed response is due to the slow thermal inertia of the oceans and the long atmospheric lifetime of greenhouse gases like $CO_2$ and methane. Committed warming highlights that current and past emissions already commit the Earth to additional warming, affecting ecosystems, sea levels, and climate patterns, and underscores the need for adaptation strategies alongside emission reductions.

> **MARKING SCHEME (3 Marks)**
> - Definition – 2 marks, effects – 1 mark

**Q.8 Why does committed warming continue even if emissions stop today?**

**Ans.** Committed warming continues due to two main factors. First, the oceans have absorbed large amounts of excess heat, which they release slowly over time, causing gradual warming of the atmosphere and climate systems. Second, long-lived greenhouse gases, especially $CO_2$, remain in the atmosphere for centuries, maintaining the enhanced greenhouse effect even if new emissions

> **MARKING SCHEME (3 Marks)**
> - Ocean heat release – 1.5 marks, long-lived greenhouse gases – 1.5 marks

studyplusplus.com (study++)

cease. Together, these factors create a lag between emission reductions and temperature stabilization. As a result, Earth's climate continues to adjust to the existing imbalance, leading to ongoing warming, sea-level rise, and other climate impacts, highlighting the importance of early mitigation and adaptation efforts.

**Q.9 What is thermal expansion and how does it raise sea levels?**

**Ans.** Thermal expansion is the process in which water increases in volume as it warms. As global temperatures rise, ocean water absorbs heat, causing water molecules to move faster and occupy more space, which expands the overall volume of the oceans. This expansion contributes significantly to rising sea levels, alongside melting glaciers and ice sheets. Thermal expansion affects coastal regions by increasing flood risks and altering ocean circulation patterns. It is one of the primary contributors to observed sea-level rise and serves as a clear example of how global warming directly impacts physical properties of natural systems, demonstrating the interconnectedness of temperature change and environmental consequences.

> **MARKING SCHEME (3 Marks)**
> - Definition – 2 marks, mechanism – 1 mark

**Q.10 How can thermal expansion be demonstrated in the lab?**

**Ans.** Thermal expansion can be demonstrated using a simple laboratory setup. Fill a flask with colored water and insert a narrow glass tube, ensuring the tube is sealed at the top. Mark the initial water level in the tube. When the flask is gently heated, the water molecules gain energy and move apart, causing the water to expand. This expansion is observed as a rise in the water level within the tube. The experiment visually illustrates how heating increases the volume of a liquid due to molecular motion, providing a clear and measurable demonstration of thermal expansion. Such experiments help students understand fundamental physical principles of global warming impacts on oceans.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark, experimental demonstration – 2 marks

# Exercise

## I Multiple Choice Questions (MCQs)

❖ **Select the correct Answer.**

1. **What is the primary difference between weather and climate?**
   A. Weather is long-term environment and climate is short-term environment.
   B. Weather is short-term changes, while climate is the long-term pattern of weather.
   C. Weather affects larger areas, while climate affects smaller regions.
   D. Climate changes daily, while weather remains constant.

2. **Which of the following is an example of a natural factor that causes climate change?**
   A. Deforestation  B. Burning of fossil fuels
   C. Volcanic activity  D. Urbanization

3. **How does ocean acidification relate to climate change?**
   A. It is caused by an increase in solar radiation.
   B. It results from increased levels of carbon dioxide in the atmosphere.
   C. It occurs due to the natural release of methane.
   D. It is caused by fluctuations in the Earth's orbit.

4. **Which of the following best describes the term "anthropogenic climate change"?**
   A. Climate change caused by natural factors such as volcanic eruptions.
   B. Climate change that occurs gradually over millions of years.
   C. Climate change results from human activities like deforestation and industrialization.
   D. Climate change due to changes in the Earth's orbit.

5. **How do Milankovitch cycles affect Earth's climate?**
   A. By increasing volcanic activity.
   B. By changing the Earth's orbit and tilt, affecting the sunlight received.
   C. By reducing the amount of greenhouse gases.
   D. By causing ocean currents to circulate heat.

6. **Which of the following impacts of climate change affects animals' migration patterns?**
   A. Increased volcanic eruptions
   B. Ocean acidification
   C. Shifting habitats due to temperature changes
   D. Urbanization

studyplusplus.com (study++)

7. **What happens to the ecosystem balance when a species goes extinct due to climate change?**
A. The balance remains unaffected.
B. It strengthens the remaining species.
C. It disrupts the entire ecosystem balance.
D. It causes new species to evolve immediately.

8. **Why is deforestation a major contributor to climate change?**
A. It releases sulfur dioxide, leading to global cooling.
B. It increases the production of solar radiation.
C. It reduces the Earth's capacity to absorb $CO_2$.
D. It has no impact on climate change.

9. **Which of the following factors directly contributes to the rise in ocean temperatures due to climate change?**
A. Increased ocean salinity
B. Melting glaciers releasing fresh water
C. Enhanced greenhouse effect trapping heat in the atmosphere
D. Reduced marine biodiversity

10. **Why does the ocean have a high heat capacity, affecting the global climate?**
A. It reflects sunlight back into space.
B. It can store large amounts of heat without a significant temperature increase.
C. It has a high salt content.
D. It has a lower density than freshwater.

11. **What role does the Atlantic Meridional Overturning Circulation (AMOC) play in ocean temperature regulation?**
A. It reduces ocean acidity.
B. It transfers warm surface water to the North Atlantic and returns cold water to the tropics.
C. It blocks the flow of ocean currents towards Europe.
D. It increases the salinity of tropical waters.

12. **What is coral bleaching, and why does it occur?**
A. It is the whitening of coral due to loss of algae, caused by higher ocean temperatures.
B. It is the natural growth process of coral reefs in warmer waters.
C. It is the formation of new coral structures due to ocean acidification.
D. It is the process of corals adapting to cooler ocean temperatures.

13. **Which oceanic region is more susceptible to ocean acidification?**
A. Equatorial regions
B. Arctic and Southern Oceans
C. Subtropical regions D. Mediterranean Sea

14. **Which phenomenon is characterized by a significant increase in ocean temperatures due to shifts in warm water toward the coast of South America?**
A. La Niña
B. Gulf Stream strengthening
C. Ocean deoxygenation D. El Niño

15. **Which of the following species is considered the first mammal to go extinct directly due to climate change?**
A. Pinta Island Tortoise B. Golden Toad
C. Bramble Cay Melomys D. Adélie Penguin

**Answer Key**

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. B | 2. C | 3. B | 4. C | 5. B | 6. C | 7. C | 8. C | 9. C | 10. B | 11. B | 12. A | 13. B | 14. B | 15. D |

## II Short Answer Questions

**Q.1 How does climate change impact ocean temperature?**

**Ans.** Climate change increases atmospheric temperatures through the enhanced greenhouse effect, and the majority of this excess heat is absorbed by the oceans. Rising sea surface temperatures affect ocean circulation, which in turn influences global weather patterns, including storms, rainfall, and wind systems. Warmer oceans also impact marine biodiversity by altering habitats, forcing species migration, and stressing coral reefs and fisheries. Additionally, higher ocean temperatures contribute to the melting of polar ice, further raising sea levels. These changes disrupt *ecosystems* and affect human communities dependent on marine resources. Overall, *ocean warming* is a key indicator of climate change and drives a cascade of environmental and ecological consequences worldwide.

**MARKING SCHEME (3 Marks)**
- Impact on temperature – 1 mark, effects on ecosystems and ice – 2 marks

studyplusplus.com (study++)

**Q.2 What is the role of the ocean's high heat capacity in climate change?**

**Ans.** The ocean's high heat capacity allows it to absorb and store vast amounts of thermal energy with relatively small temperature changes. This moderates global temperatures by reducing atmospheric warming in the short term. However, the stored heat is released slowly over decades, leading to long-term changes in ocean circulation, including currents and upwelling patterns, which influence climate globally. This delayed response contributes to committed warming, ensuring that the Earth will continue to experience temperature rise even if greenhouse gas emissions are curtailed immediately. Thus, the ocean acts as both a buffer against rapid climate change and a long-term driver of gradual global warming impacts.

> **MARKING SCHEME (3 Marks)**
> - Heat storage/moderation – 1.5 marks, long-term effects – 1.5 marks

**Q.3 Name three animal species that went extinct due to climate change.**

**Ans.** Three notable animal species that went extinct due to climate change are the Bramble Cay Melomys, the Golden Toad, and the Pinta Island Tortoise. The Bramble Cay Melomys, a rodent from a small island near Australia, was lost due to rising sea levels and flooding of its habitat. The Golden Toad, native to Costa Rica, disappeared because of altered temperature and moisture patterns in its cloud forest habitat, combined with disease. The Pinta Island Tortoise in the Galápagos Islands suffered from habitat changes and food scarcity linked to climate variations. These extinctions illustrate how climate change disrupts ecosystems, destroys habitats, and creates conditions unsuitable for species survival.

> **MARKING SCHEME (3 Marks)**
> - Bramble Cay Melomys – 1 mark, Golden Toad – 1 mark, Pinta Island Tortoise – 1 mark

**Q.4 What phenomenon results in extreme ocean heat events, like heatwaves on land?**

**Ans.** Marine heatwaves are the primary phenomenon leading to extreme ocean heat events, similar to heatwaves on land. These are periods of abnormally high sea surface temperatures that persist for days to months, often intensified by climate patterns like El Niño. Marine heatwaves disrupt marine ecosystems by stressing coral reefs, altering fish distributions, and reducing plankton productivity. They can lead to mass mortality events among marine species and have cascading effects on fisheries and food security. Such ocean heat extremes are increasing in frequency and intensity due to climate change, demonstrating how oceanic systems directly reflect atmospheric warming and impact both ecological and human communities.

> **MARKING SCHEME (3 Marks)**
> - Name of phenomenon – 1 mark, effects on ecosystems – 2 marks

**Q.5 How does ocean acidification occur?**

**Ans.** Ocean acidification occurs when atmospheric carbon dioxide dissolves in seawater, forming carbonic acid. This process lowers the pH of the ocean, making it more acidic. Acidic waters reduce the availability of carbonate ions, which are essential for marine organisms like corals, mollusks, and some plankton to form calcium carbonate shells and skeletons. The process weakens existing structures, slows growth, and can increase mortality in sensitive species. Ocean acidification also alters chemical and biological interactions within marine ecosystems, affecting biodiversity, food webs, and fisheries. It is a direct consequence of anthropogenic $CO_2$ emissions, demonstrating how human activity not only warms the planet but also changes ocean chemistry with widespread ecological impacts.

> **MARKING SCHEME (3 Marks)**
> - Cause – 1 mark, effects on organisms – 2 marks

**Q.6 What is coral bleaching?**

**Ans.** Coral bleaching is the process in which corals lose their vibrant color due to the expulsion of symbiotic algae called zooxanthellae. These algae provide corals with essential nutrients through photosynthesis and are responsible for their coloration. Elevated water temperatures, solar radiation, pollution, or other environmental stressors can trigger bleaching. Without zooxanthellae, corals lose their primary energy source, making them weak and more susceptible to disease. Prolonged stress often results in coral death, reducing reef biodiversity and altering marine ecosystems. Coral bleaching also affects human communities reliant on reefs for food, tourism, and coastal protection, highlighting the broader ecological and socio-economic impacts of ocean warming.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark, effects – 2 marks

**Q.7 Which ocean current plays a critical role in regulating the climate of the North Atlantic region?**

**Ans.** The Atlantic Meridional Overturning Circulation (AMOC) is a major ocean current system that regulates the climate of the North Atlantic region. It transports warm surface waters from the tropics northward,

studyplusplus.com

moderating temperatures in Europe, while returning colder, denser deep waters southward. This circulation influences global heat distribution, ocean nutrient cycling, and weather patterns, including rainfall and storm systems. Disruptions to the AMOC, caused by melting polar ice or freshwater input from climate change, could result in significant regional cooling or warming, sea-level changes along the Atlantic coast, and shifts in global ocean currents. AMOC is therefore vital for maintaining both regional and global climate stability.

> **MARKING SCHEME (3 Marks)**
> - Name of current – 1 mark, role/effects – 2 marks

**Q.8 Why are polar bears threatened by climate change?**

**Ans.** Polar bears are highly dependent on sea ice as a platform to hunt seals, which are their primary food source. Climate change accelerates the melting of Arctic sea ice, reducing hunting areas and forcing polar bears to swim longer distances, increasing energy expenditure. This leads to malnutrition, lower reproductive success, and higher mortality rates, particularly among cubs. Habitat loss also increases human-wildlife conflict as bears search for alternative food sources near settlements. The decline of polar bears is a direct consequence of global warming, illustrating how species that rely on specific ecosystems are extremely vulnerable to climate-induced habitat changes. Conserving sea ice habitats is critical for their survival.

> **MARKING SCHEME (3 Marks)**
> - Dependence on sea ice – 1 mark, effects of habitat loss – 2 marks

**Q.9 What impact does warmer ocean water have on oxygen levels?**

**Ans.** Warmer ocean water holds less dissolved oxygen, a phenomenon known as ocean deoxygenation. Reduced oxygen availability particularly affects deeper waters, where circulation is slower, creating hypoxic or "dead" zones where marine life struggles to survive. Oxygen depletion stresses fish, invertebrates, and plankton, reducing biodiversity and disrupting food webs. It also affects metabolic rates, growth, and reproduction in marine species. Additionally, warmer water accelerates microbial respiration, consuming more oxygen and worsening hypoxia. Ocean deoxygenation, combined with acidification and warming, represents a major threat to marine ecosystems, fisheries, and human communities that depend on seafood, highlighting the cascading impacts of climate change on ocean health.

> **MARKING SCHEME (3 Marks)**
> - Oxygen reduction – 1 mark, ecological effects – 2 marks

**Q.10 How has climate change contributed to the spread of the chytrid fungus in amphibians?**

**Ans.** Climate change has altered temperature and moisture patterns in many regions, creating favorable conditions for the chytrid fungus (*Batrachochytrium dendrobatidis*), a pathogen lethal to amphibians. Warmer temperatures and altered humidity allow the fungus to expand into highland and previously cooler habitats. The spread and increased virulence of chytrid fungus have caused widespread population declines, local extinctions, and reductions in amphibian biodiversity globally. Amphibians in tropical and subtropical regions, such as Atelopus frogs, are particularly vulnerable. Climate-driven habitat changes not only facilitate the pathogen but also weaken amphibian immune defenses, making them more susceptible. This example demonstrates the indirect but severe consequences of climate change on disease dynamics in wildlife.

> **MARKING SCHEME (3 Marks)**
> - Climate change effect – 2 marks, ecological impact – 1 mark

**Q.11 What is the primary factor causing ocean warming?**

**Ans.** The primary factor causing ocean warming is the enhanced greenhouse effect, driven by increased atmospheric concentrations of carbon dioxide, methane, and other greenhouse gases from human activities. These gases trap heat in the Earth's atmosphere, and more than 90% of the resulting excess thermal energy is absorbed by the oceans. This absorption raises sea surface temperatures, disrupts marine currents, and contributes to melting polar ice. Warmer oceans influence weather patterns, including storms and precipitation, and stress marine ecosystems by altering habitats, species distributions, and reproductive cycles. Ocean warming also drives feedback mechanisms, such as reduced albedo from melting ice, further intensifying global temperature rise.

> **MARKING SCHEME (3 Marks)**
> - Cause – 1 mark, effects on oceans – 2 marks

**Q.12 Which species went extinct in Costa Rica due to climate change-induced drought?**

**Ans.** The Golden Toad (*Incilius periglenes*), endemic to the cloud forests of Costa Rica, became extinct in the late 1980s. Prolonged droughts linked to climate change disrupted its moist forest habitat, which was critical for its breeding and survival. Changes in temperature and humidity reduced suitable egg-laying sites and contributed to disease vulnerability, particularly from pathogens like chytrid fungus. Habitat loss and environmental stress prevented the

> **MARKING SCHEME (3 Marks)**
> - Species name – 1 mark, cause/effects – 2 marks

studyplusplus.com (study++)

population from recovering, ultimately leading to extinction. The Golden Toad is widely recognized as one of the first amphibian species to vanish primarily due to climate change, highlighting the vulnerability of species with restricted ranges and specialized habitat requirements.

**Q.13 What is the relationship between sea ice and ocean temperature?**

**Ans.** Sea ice plays a critical role in regulating ocean and global temperatures by reflecting sunlight back into space, a process known as albedo. As climate change warms the planet, polar ice melts, reducing the reflective surface. Darker ocean surfaces absorb more solar energy, causing ocean temperatures to rise. This warming further accelerates ice melting, creating a positive feedback loop that amplifies global warming. Changes in ocean temperature also impact currents, weather patterns, and marine ecosystems, affecting biodiversity and coastal communities. The sea ice–temperature relationship demonstrates how small changes in polar regions can trigger cascading effects on global climate systems, emphasizing the interconnectedness of Earth's climate.

> **MARKING SCHEME (3 Marks)**
> - Relationship explanation – 1.5 marks, feedback effects – 1.5 marks

**Q.14 How does ocean acidification affect calcifying organisms?**

**Ans.** Ocean acidification occurs when dissolved carbon dioxide reacts with seawater to form carbonic acid, lowering ocean pH. This reduces the availability of carbonate ions, which are essential for calcifying organisms like corals, mollusks, and certain plankton to build calcium carbonate shells and skeletons. Acidic conditions weaken existing structures, slow growth rates, and increase mortality, threatening species survival. This disruption cascades through marine food webs, affecting predators and human industries reliant on fisheries and coral reefs. Acidification also compromises ecosystem services such as coastal protection, nutrient cycling, and biodiversity. The phenomenon highlights how human-driven $CO_2$ emissions have far-reaching impacts beyond temperature, altering ocean chemistry and biological function.

> **MARKING SCHEME (3 Marks)**
> - Mechanism – 1 mark, effects on organisms/ecosystems – 2 marks

**Q.15 Differentiate between weather and climate.**

**Ans.** Weather refers to short-term atmospheric conditions in a specific location, including temperature, humidity, precipitation, wind, and cloud cover, typically observed over hours or days. In contrast, climate describes the long-term average of weather patterns over a large region and an extended period, usually 30 years or more. Climate encompasses trends, seasonal variations, and extreme events, providing a broader understanding of regional or global atmospheric behavior. While weather can change rapidly and unpredictably, climate reflects persistent patterns shaped by geography, ocean currents, and greenhouse gas concentrations. Understanding the distinction is essential for interpreting environmental data, predicting long-term changes, and planning responses to phenomena like global warming.

> **MARKING SCHEME (3 Marks)**
> - Weather definition – 1 mark, climate definition – 1 mark, key difference – 1 mark

## III Long Questions List

**Q.1** Explain how global warming and greenhouse gases contribute to rising ocean temperatures.
**Q.2** Describe the process of ocean acidification and its impact on marine life.
**Q.3** How do El Niño and La Nñia events influence ocean temperatures and global climate patterns?
**Q.4** What role does the Atlantic Meridional Overturning Circulation (AMOC) play in ocean heat distribution, and how is it affected by climate change?
**Q.5** Discuss the impact of high ocean temperatures on coral reefs and marine biodiversity.
**Q.6** How does climate change contribute to species extinction, and provide an example?
**Q.7** Describe how climate change has affected polar ecosystems and the species that inhabit them.
**Q.8** Explain the concept of marine heatwaves and their impact on ocean ecosystems.
**Q.9** What are the challenges faced by species that become endangered due to climate changes, such as the snow *leopard?*
**Q.10** How does the melting of polar ice influence global climate patterns and ocean circulation?

studyplusplus.com (study++)